

RISC-V Board Comparison: RVV and IME Capability

Short scientific comparison of DC-ROMA K1, DC-ROMA K3, and Banana Pi K1 / BPI-F3 platforms.

Board	Best Short Use-Case Name	Core / RVV / IME Capability
DC-ROMA K1	Portable RISC-V Laptop Development	SpacemiT K1; 8-core X60 CPU; RVA22; RVV 1.0 256-bit; IME / AI-fusion engine up to 2 TOPS.
DC-ROMA K3	High-Performance RISC-V AI Computing	SpacemiT K3; 8-core X100 CPU + 8-core A100 AI-specialized cores; RVA23 + RVA22 AI; RVV 1.0 up to 1024-bit AI/vector capability; AI acceleration up to 60 TOPS.
Banana Pi K1 / BPI-F3	Embedded Industrial RISC-V Development	SpacemiT K1; 8-core X60 CPU; RVA22; RVV 1.0 256-bit; IME / AI-fusion engine up to 2 TOPS.

Scientific note: RVV indicates RISC-V vector-processing capability for parallel numerical workloads. IME / AI-fusion capability indicates matrix or AI-oriented acceleration. K1-based platforms provide 256-bit RVV 1.0 and approximately 2 TOPS AI capability, while K3 provides stronger heterogeneous compute with RVA23 CPU cores, wider AI/vector capability, and up to 60 TOPS AI acceleration.